Aror University of Art, Architecture, Design and Heritage

Faculty of Artificial Intelligence and Multimedia and Gaming

Work Sheet: 02

Q:01 Determine whether each equation represents y as a function of x.

(a) $x^2 + y^2 = 4$ (b) $x^2 - y = 9$ (c) $y = \sqrt{16 - x^2}$

(d) $y = \sqrt{x + 5}$ (e) $y = -75$ (f) $x - 1 = 0$

Q:02 Find each function value, if possible.

(a) If $f(x) = 3x - 5$ then find $f(1)$, $f(-3)$ and $f(x + 2)$.

(b) If $h(t) = -t^2 + t + 1$ then find $h(2)$, $h(-1)$ and $h(x + 1)$.

(c) If $f(y) = 3 - \sqrt{y}$ then find $f(4)$, $f(0.25)$ and $f(4x^2)$.

(d) If $q(x) = \frac{1}{x^2-9}$ then find $q(0)$, $q(3)$ and $q(y + 3)$.

(e) If $f(x) = \begin{cases} 2x + 1 & x < 0 \\ 2x + 2 & x \geq 0 \end{cases}$ then find $f(-1)$, $f(0)$ and $f(2)$.

Q:03 Find all real values of x for which $f(x) = 0$.

(a) $f(x) = x^2 - 81$ (b) $f(x) = \frac{3x-4}{5}$ (c) $f(x) = \frac{12-x^2}{8}$ (d) $f(x) = x^2 - 6x - 16$

Q:04 Find the value(s) of x for which $f(x) = g(x)$.

(a) $f(x) = x^2$, $g(x) = x + 2$ (b) $f(x) = x^2 + 2x + 1$, $g(x) = 5x + 19$

(c) . $f(x) = \sqrt{x} - 4$, $g(x) = 2 - x$

Q:05 Determine whether the function is even, odd, or neither.

(a) $f(x) = x^2 - 81$ (b) $f(x) = \frac{12-x^2}{8}$ (c) $f(x) = x^2 - 6x - 16$ (d) $f(x) = \sqrt{16 - x^2}$

Q:06 Write an equation for the function whose graph is described.

(a) The shape of $f(x) = x^2$, but shifted three units to the right and seven units down.

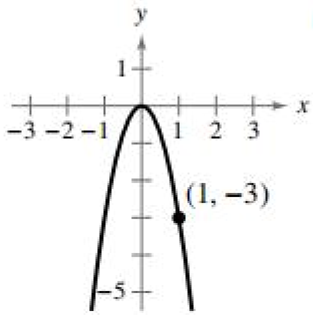
(b) The shape of $f(x) = x^3$, but shifted six units to the left, six units down, and then reflected in the y-axis.

(c) The shape of $f(x) = |x|$, but shifted 12 units up and then reflected in the x-axis.

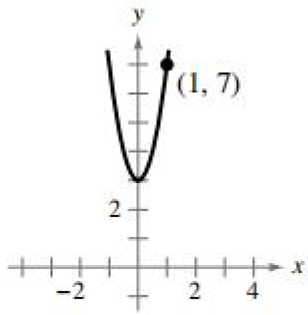
(d) The shape of $f(x) = \sqrt{x}$, but shifted six units to the left and then reflected in both the x-axis and the y-axis.

Q:07 Use the graph of $f(x) = x^2$ to write an equation for the function represented by each graph.

(a)

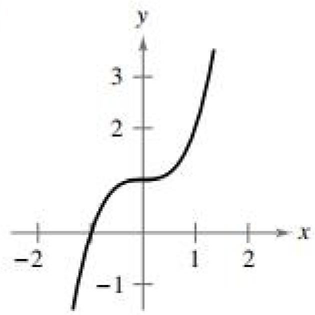


(b)

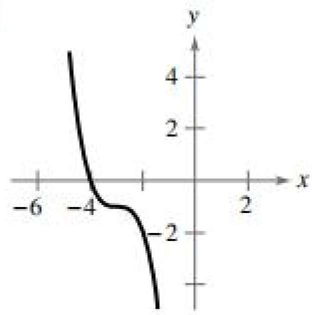


Q:08 Use the graph of $f(x) = x^3$ to write an equation for the function represented by each graph.

(a)



(b)



Q:09 Find $(f+g)(x)$, $(f-g)(x)$, $(fg)(x)$, $(f/g)(x)$, (fog)(x) and (gof)(x) for each of the following.

(a) $f(x) = x + 2$, $g(x) = x - 2$ (b) $f(x) = 2x - 5$, $g(x) = 2 - x$

(c) $f(x) = x^2$, $g(x) = 4x - 5$ (d) $f(x) = 3x + 1$, $g(x) = x^2 - 16$

Q:10 Find (a) $f \circ g$, (b) $g \circ f$, and (c) $g \circ g$, *when* $f(x) = \sqrt{x+4}$ *and* $g(x) = x^2 + 1$.

Q: 11 Find inverse function of the following functions.

(a) $f(x) = 2 - x$ (b) $f(x) = 2x - 5$, (c) $f(x) = x^2 - 81$ (d) $f(x) = \frac{3x-4}{5}$ (e) $f(x) = \sqrt{16 - x^2}$

Q:12 Use long division to divide the following.

(a) $(2x^2 + 10x + 12) \div (x + 3)$ (b) $(5x^2 - 17x - 12) \div (x - 4)$

(c) $(4x^3 - 7x^2 - 11x + 5) \div (4x + 5)$ (d) $(6x^3 - 16x^2 + 17x - 6) \div (3x - 2)$

(e) $(x^4 + 5x^3 + 6x^2 - x - 2) \div (x + 2)$ (f) $(x^3 + 4x^2 - 3x - 12) \div (x - 3)$

(g) $(3x + 2x^3 - 9 - 8x^2) \div (x^2 + 1)$

Q:13 Use synthetic division method to divide the following.

(a) $(2x^2 + 10x + 12) \div (x + 3)$ (b) $(5x^2 - 17x - 12) \div (x - 4)$

(c) $(4x^3 - 7x^2 - 11x + 5) \div (4x + 5)$ (d) $(6x^3 - 16x^2 + 17x - 6) \div (3x - 2)$

(e) $(x^4 + 5x^3 + 6x^2 - x - 2) \div (x + 2)$ (f) $(x^3 + 4x^2 - 3x - 12) \div (x - 3)$

(g) $(5x^3 + 6x + 8) \div (x + 2)$

Q:14 write the function in the form $f(x) = (x - k)q(x) + r$ for the given value of k, and demonstrate that $f(k) = r$.

(a) $f(x) = x^3 - x^2 - 10x + 7$, $k = 3$ (b) $f(x) = x^3 - 4x^2 - 10x + 8$, $k = -2$

(c) $f(x) = 15x^4 + 10x^3 - 6x^2 + 14$, $k = -2/3$ (d) $f(x) = 10x^3 - 22x^2 - 3x + 4$, $k = 1/5$

(e) $f(x) = -4x^3 + 6x^2 + 12x + 4$, $k = 1 - \sqrt{3}$

Q:15 use synthetic division to show that x is a solution of the third-degree polynomial equation, and use the result to factor the polynomial completely. List all real solutions of the equation.

(a) $X^3 + 6x^2 + 11x + 6 = 0$, $x = -3$ (b) $x^3 - 52x - 96 = 0$, $x = -6$

(c) $2x^3 - 15x^2 + 27x - 10 = 0$, $x = 1/2$ (d) $48x^3 - 80x^2 + 41x - 6 = 0$, $x = 2/3$